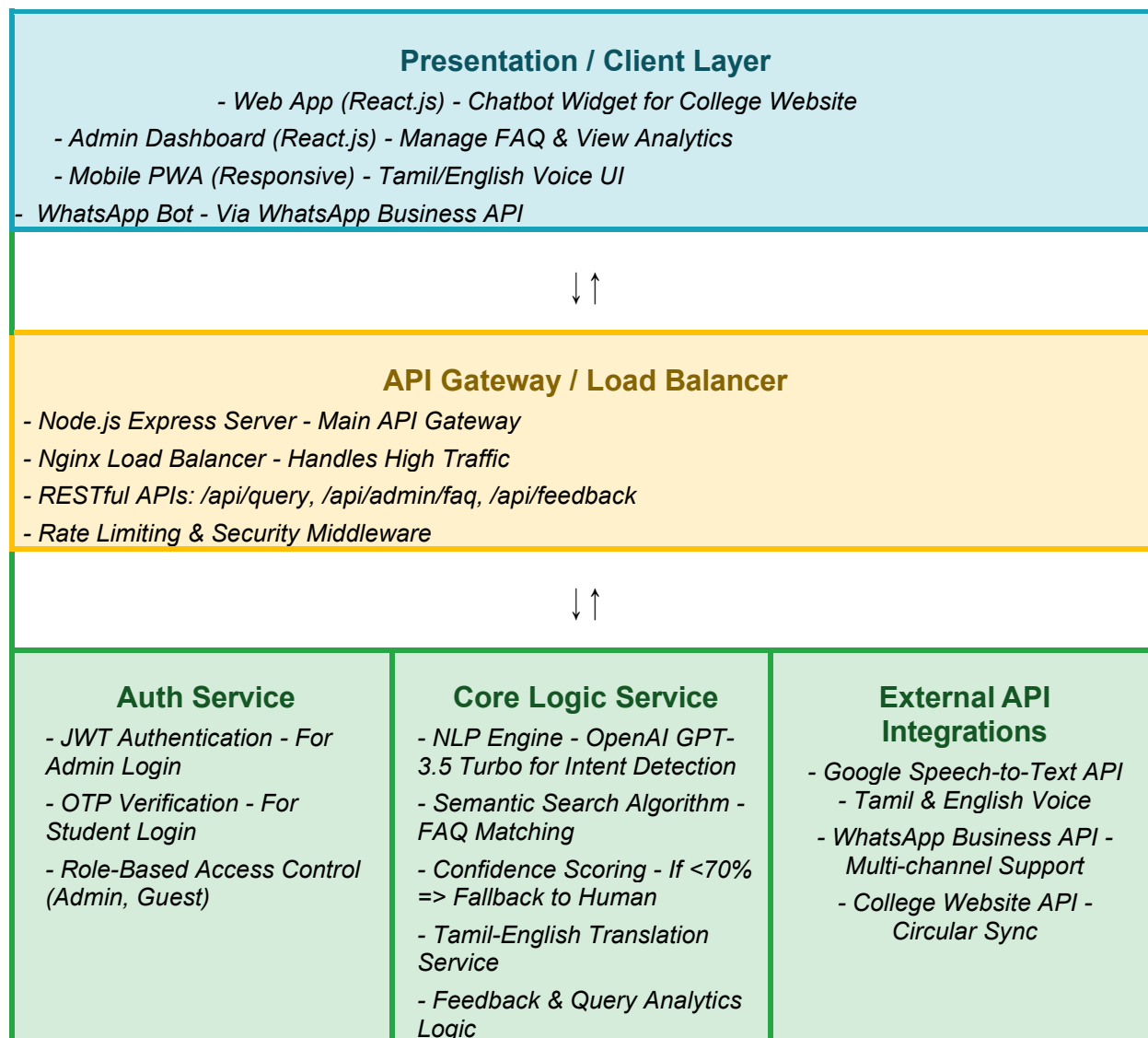


Solution Architecture

Date	29 September 2026
Team ID	SWTID-2026-7674
Project Name	AI FAQ ASSISTANT
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.





Data / Storage Layer

- D1: MySQL Database - FAQ Knowledge Base (Verified Q&A)
- D2: MongoDB - User Query Logs & Unanswered Queries
 - D3: Redis Cache - For Fast Response (<2 sec)
 - AWS S3 Bucket - Daily Backup & File Storage
- Vector DB (Pinecone) - For Semantic Search Embeddings

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>This is the user interface layer where students/parents interact with chatbot. It displays answers, takes text/voice input (Tamil & English), and provides Admin Dashboard to manage FAQs. Handles 24/7 user interaction via website & WhatsApp.</i>	<i>React.js, HTML/CSS, JavaScript, PWA, WhatsApp Business API Widget, Responsive UI</i>
<i>API Gateway</i>	<i>Acts as single entry point for all requests. Routes requests to correct microservice, handles load balancing during admission season, security, rate limiting, and JWT validation. Reduces direct load on services.</i>	<i>Node.js Express Server, Nginx Load Balancer, REST APIs, JWT Middleware</i>
<i>Microservices</i>	<i>Auth Service: Manages admin login with JWT & OTP, role-based access. Core Logic Service: Heart of system - processes query using NLP, does semantic search on FAQ DB, checks confidence score, translates Tamil to English and vice-versa, decides fallback to human.</i>	<i>OpenAI GPT-3.5 Turbo API, Python NLP (spaCy), Node.js, Google Speech-to-Text API, Tamil Translation API]</i>
<i>Database/Storage Layer</i>	<i>Stores all data permanently and for fast retrieval. MySQL stores verified FAQs, MongoDB stores query logs & unanswered questions for improvement, Redis cache for instant <2 sec reply, S3 for backup. Vector DB stores embeddings for semantic search.</i>	<i>MySQL, MongoDB, Redis Cache, AWS S3, Pinecone Vector DB</i>